

Software Requirements Specification (SRS)

Community Tool & Equipment Library — Problem 52

1. Introduction

This SRS defines the requirements for the Community Tool & Equipment Library system. It organizes the submitted functional and non-functional requirements into a software specification for the system.

2. Purpose

The system supports community members and library custodians in finding, reserving, checking out, returning, and managing shared tools and equipment, including deposits, late fees, damage assessment, and notifications.

3. Scope

- Tool inventory search and filtering by category, availability, and depot locker location.
- Tool reservation and checkout with required deposit processing.
- Tool return processing, condition recording, damage assessment, and late-fee calculation.
- Due-date and overdue notifications.
- Security-deposit and loan-duration rules.
- Real-time physical availability visibility across depot lockers.
- Responsive access on standard mobile devices.

4. Actors

- Community Member — searches, reserves, checks out, returns tools, and receives notifications.
- Library Custodian — processes returns, records condition, manages damage and late fees.
- System/Administrative functions — enforce deposit, loan-duration, inventory and notification rules.

5. Functional Requirements

ID	Requirement
FR-001	Calculate required security deposits, enforce loan-duration limits, and calculate late-return fees.
FR-002	Allow members to search and filter tool inventory by category, availability, and depot locker location.
FR-003	Allow members to reserve and check out an available tool after required deposit payment is authorized.
FR-004	Allow custodians to process returns, record condition, and log damage assessment and/or late fees.
FR-005	Automatically send a reminder 24 hours before the due date and an overdue notification after the due date.

6. Non-Functional Requirements

ID	Requirement
NFR-001	Tool inventory search must display real-time physical availability across neighborhood depot lockers.
NFR-002	Core member-facing screens (search, reserve, return) should render within 3 seconds on a 4G connection for 95% of measures.

7. Acceptance Criteria

- Checkout is confirmed only after deposit authorization and a due date is generated.
- Unavailable or already checked-out tools must not appear as currently available search results.
- A returned tool must no longer remain in checked-out status.
- Damage and late conditions must be reflected in the appropriate records/fees.
- Notification records must be created and delivered near the scheduled trigger time.
- Mobile core screens must meet the specified rendering target.

8. System Constraints and Assumptions

- The system depends on current inventory/depot availability data.
- Payment processing requires a payment service or gateway.
- Notifications require an email/SMS/push delivery mechanism.
- Performance measurements depend on the test environment and simulated network conditions.

9. Traceability

FR-001 through FR-005 and NFR-001 through NFR-002 should be traced to design components, implementation tasks, and test cases in the project RTM and later testing evidence.